IPCON® CoPA HS

High Speed Polyamide 6/66 copolymer Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	365 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or PC (max heat resistance: 130 °C)
Spool Size	Outer cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm
Recommendation for body protection during use	No special requirements

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.14 g/cm³
Shrinkage Rate*	Specially-defined	0.28% - 0.49%
Surface Hardness	Shore hardness	84D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	2.98%
Melt Index	280 °C, 2.16 kg	16.3 g/10 min
Melting Temperature	DSC, 10 °C/min	237 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	Not annealed: 87 °C Annealed: 158 °C*
Heat Deflection Temperature	ISO 75, 0.45 MPa	Not annealed: 150 °C Annealed: 152 °C*
Heat Deflection Temperature	ISO 75, 1.80 MPa	Not annealed: 65 °C Annealed: 70 °C*
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	60 - 80 °C (dry state) 50 - 60 °C (wet state) (Related to load, size, shape, wall thickness, rate of water content, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*	
		Dry State	Wet State (the rate of water content is 2.98%)
Tensile Strength XY	ISO 527, GB/T 1040	63 MPa	31 MPa
Tensile Strength Z	ISO 527, GB/T 1040	31 MPa	19 MPa
Young's Modulus XY	ISO 527, GB/T 1040	2653 MPa	793 MPa
Young's Modulus Z	ISO 527, GB/T 1040	1782 MPa	538 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	12.6%	45.3%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	13.8%	271.6%
Flexural Strength XY	ISO 178, GB/T 9341	113 MPa	42 MPa
Flexural Strength Z	ISO 178, GB/T 9341	65 MPa	31 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	3068 MPa	1036 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2385 MPa	691 MPa
Impact Strength XY	ISO 179, GB/T 1043	64.3 kJ/m²	124.1 kJ/m² (No Break)
Impact Strength Z	ISO 179, GB/T 1043	13.3 kJ/m²	24.7 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 295 °C, bed temperature: 100 °C, printing speed: printing speed: 100 mm/s) and IPCON PAHT GF (red) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The XY specimens' annealing parameter is: 80 °C, 4 h → 100 °C, 8 h. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Smooth surface, low friction coefficient, resistant to wear, resistant to oil and general chemical reagents, high strength, rigidity and toughness in dry state, ultra-high toughness in wet state, ultra-low warpage, suitable for AMS or CFS printing, 4 basic colors of black, white, gray and red.
2. Gears, slide rails, bushings and other parts that require wear resistance, mechanical arms, drone brackets, toy racing cars, motorcycle fenders and other parts that require a balance of strength, stiffness and toughness.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 120 °C, 8 - 14 h (Blast drying oven)
Printing Temperature and Humidity	≤ 70 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PA
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Compatible
Compatible Nozzle Material	Any common material, such as brass, stainless steel, hardened steel, diamond, tungsten carbide, etc

Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	270 - 300 °C
Bed Temperature	90 - 110 °C
Fan Speed	0% - 40%
Max Print Speed	300 mm/s

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